0.0.1 Metric Space

Definition 0.0.1. Let X be a set, and $d : X \times X \to \mathbb{R}$ is **metric** on X if:

D1. For any $x, y \in X$, $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$.

D2. For any $x, y \in X$, $d(x, y) = d(y, x)$.

D3. For any $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.

We say that (X, d) is a **Metric Space**.

Definition 0.0.2. Let (X, d) be a Metric Space.

1. **Open ball** of X: $B_r(x_0) \overset{\text{def}}{=} \{x \in X \mid d(x, x_0) < r\}$.
2. **Closed ball** of X: $\bar{B}_r(x_0) \overset{\text{def}}{=} \{x \in X \mid d(x, x_0) \leq r\}$.
3. **Sphere** of X: $S_r(x_0) \overset{\text{def}}{=} \{x \in X \mid d(x, x_0) = r\}$.

Lemma 0.0.3. Let X be a set and $d : X \times X \to \mathbb{R}$ be Metric of X. Then,

$$\beta_d \overset{\text{def}}{=} \{B_r(x) \mid r \in (0, \infty),\ \ x \in X\}$$

be a basis of X.

Proof.
B1. Let $x \in X$ be given. Then,$B_r(x) \in \beta_d$ satisfies $x \in B_r$.
B2. Let two balls $B_{r_1}(x), B_{r_2}(y) \in \beta_d$ are given. And, for any $z \in B_{r_1}(x) \cap B_{r_2}(y)$, set $r_3 = \min\{r_1 - d(x, z), r_2 - d(y, z)\}$. Then, we obtain: for any $w \in B_{r_3}(z)$, i.e., $d(w, z) < r_3$,

$$d(x, w) \leq d(x, z) + d(z, w) < d(x, z) + r_3 \leq d(x, z) + (r_1 - d(x, z)) = r_1$$
$$d(y, w) \leq d(y, z) + d(z, w) < d(y, z) + r_3 \leq d(y, z) + (r_2 - d(y, z)) = r_2$$

thus, $w \in B_{r_1}(x) \cap B_{r_2}(y)$, we get $B_{r_3}(z) \subset B_{r_1}(x) \cap B_{r_2}(y)$ and $z \in B_{r_3}$ is trivial. □

Definition 0.0.4. Let X be a set, and $d : X \times X \to \mathbb{R}$ is metric on X.
We say that $(X, \mathcal{T}_d)$ is **Metric Space** where $\mathcal{T}_d \overset{\text{def}}{=} \mathcal{T}_{\beta_d}$. Clearly,

$$\mathcal{T}_d = \{U \subset X \mid \forall x \in U,\ \exists B \in \beta_d \ \text{ s.t } \ x \in B \subset U\}$$

Lemma 0.0.5. Let Metric Space (X, d), and $U \subset X$, TFAE:

a) $U \in \mathcal{T}_d$.

b) For any $x \in U$, there exists $r > 0$ such that $x \in B_r(x) \subset U$.

Proof.
$2. \implies 1.$ is trivial by $B_r(x) \in \beta_d$.
To show that $1. \implies 2.$,
Let $x \in U$ be given. Since $U \in \mathcal{T}_d$, there exists $B_\delta(y) \in \beta_d$ such that $x \in B_\delta(y) \subset U$. Let $r = \delta - d(x, y)$.
Then, $x \in B_r(x) \subset B_\delta(y)$ because for any $w \in B_r(x)$,

$$d(w, y) \leq d(w, x) + d(x, y) < \delta - d(x, y) + d(x, y) = \delta$$

□

Lemma 0.0.6. Let X be a set, $d_1, d_2 : X \times X \to \mathbb{R}$ be metric. TFAE:

1. $\mathcal{T}_{d_1} = \mathcal{T}_{d_2}$.
2. For any $x \in X$, $\varepsilon > 0$, there exists $\delta > 0$ such that $B_{d_2}(x;\delta) \subset B_{d_1}(x;\varepsilon)$.

Definition 0.0.7. Let $(X, \mathcal{T})$. Define T_3 **Condition** on the space:

$$\forall x, y \in X,\ \ x \neq y \implies \exists U, V \in \mathcal{T} \ \text{ s.t } \ x \in U,\ \ y \in V,\ \ U \cap V = \emptyset$$

If $(X, \mathcal{T})$ satisfies T_3 **Condition**, we say that this space is **Hausdorff Space**.

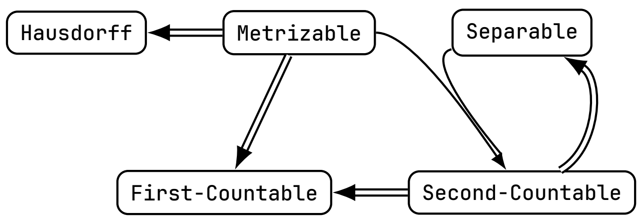
Definition 0.0.8. Topological Space $(X, \mathcal{T})$ is **Metrizable** if:
there is a metric $d : X \times X \to \mathbb{R}$ such that $\mathcal{T} = \mathcal{T}_d$.

Definition 0.0.9. Let $\mathbb{R}^n$ be given. Define p**-norm** of $\mathbb{R}^n$ is metric on $\mathbb{R}$:

$$d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R} : (\mathbf{x}, \mathbf{y}) \mapsto \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}}, \quad (\mathbf{x} = (x_1, \ldots, x_n),\ \ \mathbf{y} = (y_1, \ldots, y_n))$$

where $p \in [1, \infty]$, p-norm be a **Metric** by **Minkowski inequality**.

0.0.2 Metrizable Theorem Diagram



Topology for $\mathbb{R}$: $\mathcal{T}_{id} \subset \mathcal{T}_f \subset \mathcal{T}_u \subset \mathcal{T}_l \subset \mathcal{T}_d,\ \mathcal{T}_f \subset \mathcal{T}_c$						
Topology	$\mathcal{T}_{id}$	$\mathcal{T}_f$	$\mathcal{T}_u$	$\mathcal{T}_l$	$\mathcal{T}_d$	$\mathcal{T}_c$
First-Countable	Yes	No	Yes	Yes	Yes	No
Second-Countable	Yes	No	Yes	No	No	No
Hausdorff	No	No	Yes	Yes	Yes	No
Metrizable	No	No	Yes	No	Yes	No
Separable	Yes	Yes	Yes	Yes	No	No
Compactness	Yes	Yes	No	No	No	Lindelof

Let a Topological Space $(X, \mathcal{T})$ be given. For X, the following theorems hold:

> **Theorem 0.0.10.** If Metrizable, then Hausdorff.

Proof. Let $(X, \mathcal{T})$ be a metrizable topological space. Then, there is a metric $d : X \times X \to \mathbb{R}$.
Let distinct two point $x, y \in X$ be given. And, put $\varepsilon = \frac{d(x,y)}{2} > 0$.
Now, $x \in B_d(x; \varepsilon)$ and $y \in B_d(y; \varepsilon)$ but there intersection is empty set because:
Choose $p \in B_d(x; \varepsilon) \cap B_d(y; \varepsilon)$. Then, $d(x, p), d(y, p) < \varepsilon$.
But, this implies that

$$d(x, y) \leq d(x, p) + d(p, y) < 2\varepsilon = d(x, y)$$

Thus, contradiction. Consequently, given topological space satisfies Hausdorff property. □

> **Lemma 0.0.11.** Let (X, d) be a Metric Space.
> For any $U \in \mathcal{T}_d$, $x \in U$, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.

Proof. Let $U \in \mathcal{T}_d$ be given.
Then, there exist $y \in X$ and $\varepsilon > 0$ such that $x \in B_d(y, \varepsilon) \subset U$. For $\delta = \varepsilon - d(x, y) > 0$, $B_d(x, \delta) \subset B_d(y, \varepsilon)$ because:

$$\begin{aligned} p \in B_d(x, \delta) &\iff d(p, x) < \delta \\ &\implies d(p, y) \leq d(p, x) + d(x, y) < \delta + d(x, y) = \varepsilon - d(x, y) + d(x, p) = \varepsilon \\ &\iff p \in B_d(y, \varepsilon) \end{aligned}$$

□

> **Theorem 0.0.12.** If Metrizable, then First-Countable.

Proof. Let $(X, \mathcal{T})$ be a metrizable topological space, and d be a metric that induces $\mathcal{T}$.
For any $x \in X$, define

$$\beta_x \overset{\text{def}}{=} \left\{ B_{\frac{1}{n}}(x) \,\middle|\, n \in \mathbb{N} \right\}$$

Trivially, β_x is countable set. Want To Show: β_x be a local base of x.
LB1) is clear.
LB2) Let $U \in \mathcal{T}$ with $x \in U$ be given. Since above lemma, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.
Let $n \in \mathbb{N}$ be a number such that $1 < \delta n$, then $B_d\left(x, \frac{1}{n}\right) \in \beta_x$ satisfies

$$x \in B_d\left(x, \frac{1}{n}\right) \subset B_d(x, \delta) \subset U$$

□

> **Theorem 0.0.13.** If Metrizable and Separable, then Second-Countable.

Proof. Since X is Separable, let $A = \{x_i \mid i \in \mathbb{N}\}$ be a countable dense subset of X.
Let $d : X \times X \to \mathbb{R}$ be a metric on X that induces $\mathcal{T}$. Define

$$\beta \overset{\text{def}}{=} \left\{ B_d\left(x_i, \frac{1}{n}\right) \,\middle|\, i, n \in \mathbb{N} \right\}$$

Clearly, β is Countable and $\beta \subset \mathcal{T}$.
Proof 1. Using B1 and B2
B1) Let $x \in X$. Since $X = \bar{A}$, $x \in A$ or $x \in A'$.
If $x \in A$, then $x = x_i$ for some $i \in \mathbb{N}$, thus $x \in B_d(x_i, 1)$.
If $x \notin A$, then $x \in A'$ and $x \notin A$.
Let $U \in \mathcal{T}$ with $x \in U$ be given. Then, from above lemma, there exists $r > 0$ such that $x \in B_d(x, r) \subset U$.
Set $n \in \mathbb{N}$ such that $1 < nr$ and $\delta = \frac{1}{n}$, then $B_d(x, \delta) \in \beta$ is open and

$$A \cap (B_d(x, \delta) \setminus \{x\}) = (A \setminus \{x\}) \cap B_d(x, \delta) = A \cap B_d(x, \delta) \neq \emptyset$$

This implies that for some $i \in \mathbb{N}$, $d(x_i, x) < \frac{1}{n}$. Now,

$$x \in B_d(x_i, \delta) \in \beta$$

B2) It is proved in the same way as the previous discussion.

Proof 2. Direct to Definition
Enough to Show that: For any $U \in \mathcal{T}$, there exists $\mathcal{C} \subset \beta$ such that

$$U = \bigcup_{u \in \mathcal{C}} u$$

Let $U \in \mathcal{T}$ be given. Since $\mathcal{T}$ is consisted by open balls induced a metric d,
for any $x \in U$, there exists $r > 0$ such that $B_d(x, r) \subset U$ since above lemma.
Set $\delta_x = \frac{1}{2n}$ such that $1 < nr$. Since A is dense in X,

$$A \cap B_d\left(x, \delta_x\right) \neq \emptyset$$

i.e., for some $i \in \mathbb{N}$, $d(x, x_i) < \delta_x$. Thus,

$$x \in B_d\left(x_i, \delta_x\right) \subset B_d(x, 2\delta_x) \subset B_d(x, r) \subset U$$

Consequently, we shown that:

$$\forall x \in U, \ \exists B_x \in \beta \ \text{ s.t } \ x \in B_x \subset U$$

Now, Construct given $U \in \mathcal{T}$ as: Since **Axiom of Choice**, Define $\mathcal{C} \overset{\text{def}}{=} \bigcup_{x \in U} \{B_x \in \beta \mid x \in B_x \subset U\}$. Then,

$$U = \bigcup_{u \in \mathcal{C}} u$$

□

> **Theorem 0.0.14.** If Second-Countable, then Separable.

Proof. Let $\beta = \{B_i \mid i \in \mathbb{N}, \ B_i \neq \emptyset\}$ be a Countable Basis of X.
By the **Axiom of Choice**, we can construct a countable set $A = \{x_i \mid i \in \mathbb{N}, \ x_i \in B_i\}$. WTS: $\bar{A} \supset X$.
Let $x \in X$ and $U \in \mathcal{T}$ with $x \in U$ be given arbitrarily.
Since β is basis of $\mathcal{T}$, $x \in B_i \subset U$ for some $B_i \in \beta$. Thus, $x_i \in U$, so we obtain $A \cap U \neq \emptyset$. Therefore, $x \in \bar{A}$. □

> **Theorem 0.0.15.** If Second-Countable, then First-Countable.

Proof. Trivial. □

Proposition 0.0.16. Lower limit Topology on $\mathbb{R}$ is Not Second-Countable.

Proof. Suppose that $(\mathbb{R}, \mathcal{T}_l)$ has a countable basis β.
Then, for any $x \in \mathbb{R}$, there exists a $B_x \in \beta$ s.t. $x \in B_x \subseteq [x, x+1)$.
Clearly, if $x < y$, $B_x \neq B_y$. (Because $x \in B_x$ but $x \neq B_y$.) Now, the map $\mathbb{R} \to \beta; x \mapsto B_x$ is injective.
Thus $\#(\beta) > \aleph_0$, which is contradiction. □

Proposition 0.0.17. Co-Finite Topology on $\mathbb{R}$ is Not Second-Countable.

Proof. Let β be a countable basis for $\mathcal{T}_f$ not containing $\emptyset$.
(If β has $\emptyset$, $\beta \setminus \{\emptyset\}$ is still a basis which generated same Topology).
Now, $\mathbb{R} \neq \bigcup_{B \in \beta} B$ because $\mathbb{R} \setminus B$ is finite and β is countable. Let $x \in \mathbb{R} \setminus \left(\bigcup_{B \in \beta} B\right)$. Then, $\mathbb{R} \setminus \{x\}$ is open of $\mathcal{T}_f$.
But, $x \in \mathbb{R} \setminus \bigcup_{B \in \beta} B$ implies that $\forall B \in \beta, x \notin B$.
By definition, for any $p \in \mathbb{R} \setminus \{x\}$, there exists $B \in \beta$ s.t.

$$p \in B \subseteq \mathbb{R} \setminus \{x\}$$

which means $\{x\} \subseteq B^c$. Contradiction. □

Note: Similarly, Co-countable space on $\mathbb{R}$ is Not Second-countable.